# Who Benefits from the Minimum Wage?

## A Product Market Mechanism

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# Motivation

**Robust empirical response to $\uparrow w_{\text{min}}$ across countries and settings:**

- Workers reallocate from low- to high-productivity firms.
  Rao and Risch (2026); Dustmann et al. (2022); Engbom and Moser (2022); Haelbig et al. (2023).
- High-productivity firms become **more** productive/profitable; low-productivity firms become **less**.
  Rao and Risch (2026); Du and Wang (2020); Mayneris et al. (2018); Draca et al. (2011); Luca and Luca (2018).

**The puzzle:**

Why do firms already paying **above** the minimum wage expand and become more productive?

# Outline

1. Current explanations and their limitations.
2. Our mechanism.
3. Testable predictions.

# Current Explanations: Labor Market Stories

**The literature attributes these patterns to:**

1. **Monopsonistic labor market** (Berger et al. 2025; Dustmann et al. 2022; Haanwinckel 2025)
   $w_{\text{min}}$ cuts markdown of low productivity firms → they ration employment → workers reallocate up the productivity ladder
2. **Search frictions & congestion** (Dustmann et al. 2022; Engbom and Moser 2022; Drechsel-Grau 2023)
   Low-productivity firms cut vacancies → reduced congestion → easier recruiting for high-productivity firms
3. **Managerial slack** (Hau et al. 2020; Mayneris et al. 2018; Riley and Rosazza Bondibene 2017)
   Wage shock forces organizational improvements: investment in training, technology and reorganization.

All fundamentally **labor market** mechanisms

# Problems with Labor Market Stories (1/2)

## 1. Monopsony & Search Frictions

Both require market frictions that are hard to sustain in low-skill labor markets:

- **Thick markets:** many similar employers, substitutable workers — the "few alternatives" premise is implausible.
- **Standardized jobs:** low specialization $\Rightarrow$ fast, cheap matching $\Rightarrow$ search frictions are mechanically small.
- **Empirically:** near-complete price pass-through (Allegretto and Reich 2018; Harasztosi and Lindner 2019); $V \approx 1.04$ in US restaurants (Aaronson and French 2007).

Low-skill labor is precisely where competitive markets work best — frictions are mechanically smallest.

# Problems with Labor Market Stories (2/2)

2. **Managerial Slack / Effort**
   - **Issue 1:** Why does it require a wage shock? If productivity improvements were available, profit-maximizing firms would already implement them.
   - **Issue 2:** The mechanism is **one-sided** — predicts improvements only for firms that received the shock. Our puzzle is the other side: **unaffected** high-productivity firms also expand. Draca et al. 2011; Haelbig et al. 2023
   - **Issue 3:** $w_{\text{min}}$ directly affects only low-wage workers, yet claimed gains are firm-wide — requiring large positive spillovers from low-skill workers to the entire organization.

# This Paper: Product Market Competition

**Unlike labor market stories, the mechanism operates through relative prices — within the same industry:**

1. **Vanishing burden:** High-productivity firms already pay above the floor — their direct cost increase is negligible.
2. **Uneven cost increase:** All firms' costs rise, but more productive firms' costs rise less $\Rightarrow$ their relative marginal cost $\frac{\psi\omega}{A(p)}$ falls $\Rightarrow$ their **relative price** falls.
3. **Demand reallocation:** Lower relative price $\rightarrow$ market share and profit rise for productive firms; low-productivity firms contract.
   - The resulting revenue and market share gains appear as productivity increases in standard measures — competitive position, not efficiency.

→ Theoretical environment.

# Why Existing Frameworks Fall Short

A small section of the literature attributes these effects to product market mechanisms (Rao and Risch 2026; Williamson 1968; Haucap et al. 2001). These papers follow the RRC/IO tradition: Cournot competition with **homogeneous goods**.

With homogeneous goods, all firms sell at a **single market price** $p^*$ — consumers cannot substitute toward a specific firm because its relative price fell. Demand reallocation through differential price changes is **not expressible** in this framework.

**One exception:** The one empirical paper implicitly requiring differentiated goods — Harasztosi and Lindner (2019) — imposes a **proportional** cost shock: all prices rise together, and no firm's relative price changes.

# Testable Predictions

**If the product market mechanism operates:**

- **Relative prices:** least productive firms raise prices more after $\uparrow w_{\text{min}}$; productive firms raise prices less.
- **Market share:** productive firms gain market share within the industry; unproductive firms lose.
- **Employment:** follows output — productive firms hire more, not because of labor market pressures, but because demand shifted.

**These predictions are untested.**

Restaurant-sector papers are the only within-industry studies, but assume homogeneous goods or putty-clay technology and draw no conclusions about productivity redistribution.

# Conclusion

1. Minimum wage generates a **product market** effect via the price aggregator: heterogeneous cost shock → relative price shift → demand reallocation within the industry.
2. High-productivity firms gain — not through labor market channels, but because they already pay above the floor: all costs rise, theirs least, so their **relative prices fall**.
3. Measured productivity gains are **competitive position effects**, not genuine efficiency improvements.
4. The right test is **within-industry**: relative prices and market shares. These predictions are currently untested.

# Appendix

**Model**

- Regulation class & Amazon example.
- Environment and equilibrium.
- When can higher costs help a firm?
- Theorem 1: shape-changing regulation.
- CES demand with Pareto costs.
- Why measured productivity rises.

**Literature**

- Detailed literature summary.

# The regulation class we study

**Target: industry-wide rules that raise marginal costs, but whose marginal burden collapses for large/productive firms.**

**Amazon example:**

- 2012: Acquires Kiva Systems (now Amazon Robotics, automation arm).
- 2016: Starts integrating robotics in fulfillment centers (min wage workers).
- **2018: Push for doubling the federal min wage.**
- 2022: Massive layoffs start (mainly in fulfillment centers).
- 2025: Internal leaked documents suggest that automation is going to save them 600,000 jobs over the next decade.

# Environment

- Continuum of differentiated goods $\omega \in (0, 1)$, in static closed economy.
- Consumers represented by a homothetic demand system with price aggregator $A(\mathbf{p})$:

$$x_\omega = x_\omega \left( \frac{p_\omega}{A(\mathbf{p})} \right)$$

- Heterogeneous firms in monopolistic competition with marginal cost $\psi_\omega > 0$ maximize:

$$\pi_\omega = (p_\omega - \psi_\omega) x_\omega$$

# Environment

Using HSA demand systems (Matsuyama & Ushchev, 2020), you can rewrite the firms maximization problem the following way:

$$\pi_\omega = (p_\omega - \psi_\omega) x_\omega \left( \frac{p_\omega}{A(\mathbf{p})} \right) = \left( 1 - \frac{\psi_\omega / A(\mathbf{p})}{z_\omega} \right) Y s_\omega \left( z_\omega \right)$$

Where $s_\omega(\cdot)$ is the share function, $Y$ is expenditure, and $A(\cdot)$ a homothetic aggregator. Finally $z_\omega \equiv \frac{p_\omega}{A(\mathbf{p})}$ is the relative price.

**Key Implication:** Maximizing over price levels $p_\omega$ is equivalent to maximizing over relative prices $z_\omega$, and equilibrium profit and shares are decreasing in the **relative marginal cost** $\boldsymbol{\frac{\psi_\omega}{A(\mathbf{p})}}$: $\frac{\partial \pi_\omega^*}{\partial \psi_\omega / A(\mathbf{p})} < 0$.

# When can higher costs help a firm?

From $\frac{\partial \pi^*_\omega}{\partial \psi_\omega / A(\mathbf{p})} < 0$ and $\mathbf{p}^* = \boldsymbol{\mu\psi}$, firm $\omega$ benefits from a regulatory change in marginal costs whenever its **relative marginal cost falls**:

$$\frac{\psi^R_\omega}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

**even if** its own marginal cost rises, $\psi^R_\omega > \psi_\omega$.

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# Regulations

Suppose marginal costs come from a family of continuous parametric CDFs, $\psi \sim G \in \mathcal{G}$.

**Def. Regulation:**

A regulation changes marginal costs to $\psi^R \sim G^R$, with $G, G^R \in \mathcal{G}$.

**Why parameterize?** It lets us classify regulations by the way they deform the distribution: **scale**, **location**, or **shape**.

**Thm 1:** Regulation that changes the shape parameter will lead to a decrease in relative cost for the most productive, and increase their market share, profit and price levels.

E.g., in Pareto costs: $G(\psi) = 1 - \left(\frac{\psi_0}{\psi}\right)^{\kappa}$, $\kappa$ is the shape parameter.

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# Example: CES demand with Pareto costs

The relative marginal cost for firm $\omega \in (0,1)$ would be:

$$\frac{\psi_\omega}{\mu A(\boldsymbol{\psi})} = \frac{(1-\omega)^{-\frac{1}{\boldsymbol{\kappa}}}}{\mu \left[\frac{\boldsymbol{\kappa}}{\boldsymbol{\kappa}+\sigma-1}\right]^{\frac{1}{1-\sigma}}}$$

Which implies that any decrease from any initial $\kappa$ would benefit the most productive firms. If we assume that a regulation marginally decreases $\kappa$ (Brazil's SPED), then every firm $\omega \in (0,\varepsilon)$ will benefit, with:

$$\varepsilon = 1 - \exp\left(-\frac{\kappa}{\kappa+\sigma-1}\right)$$

# Why Measured Productivity Rises

- **Labor productivity** $\equiv$ (remuneration + **profits**) / employment.
  Riley and Rosazza Bondibene (2017). Profit is included.
- High productivity firms gain market share $\Rightarrow$ revenues rise $\Rightarrow$ profits rise $\Rightarrow$ measured productivity rises.
- **This is not** a genuine efficiency gain — it is a **competitive position** gain.
- Same logic applies to TFP: higher output at similar inputs $\Rightarrow$ positive Solow residual. Scale effects and markup changes contaminate the TFP residual.

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# Detailed Literature Summary (1/2)

- **Dustmann et al. (2022):** Germany. MW $\rightarrow$ workers reallocate to high-wage firms. Attributed to monopsonistic competition. Uses cross-industry variation.
- **Engbom and Moser (2022):** Brazil. High-productivity firms increase recruiting. Attributed to vacancy congestion.
- **Rao and Risch (2026):** US pass-through firms. Revenues $+4.3\%$, profits unchanged, positive entry selection. Uses Cournot framework in appendix. Cross-industry treatment definition.
- **Harasztosi and Lindner (2019):** Hungary. Prices rise, employment unchanged. Market and firm-level elasticity distinction — partial product market framework. No firm heterogeneity test.

# Detailed Literature Summary (2/2)

- **Aaronson and French (2007):** US restaurants. Pass-through $\approx 7\%$ per $10\%$ MW — rules out monopsony. Representative firm model; homogeneous product.
- **Luca and Luca (2018):** Yelp. Low-rated restaurants exit disproportionately. Explicit Williamson (1968) citation — intra-industry evidence consistent with our Cor. 1.
- **Riley and Rosazza Bondibene (2017):** UK. Labor productivity $\equiv$ (remuneration + profits)/employment. Productivity gains include competitive position gains.

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# References I


Aaronson, D. and E. French (2007): "Product Market Evidence on the Employment Effects of the Minimum Wage," *Journal of Labor Economics*, 25, 167–200.

Allegretto, S. and M. Reich (2018): "Are Local Minimum Wages Absorbed by Price Increases? Estimates from Internet-Based Restaurant Menus," *ILR Review*, 71, 35–63.

Berger, D., K. Herkenhoff, and S. Mongey (2025): "Minimum Wages, Efficiency, and Welfare," *Econometrica*, 93, 265–301.

Draca, M., S. Machin, and J. Van Reenen (2011): "Minimum Wages and Firm Profitability," *American Economic Journal: Applied Economics*, 3, 129–151.

Drechsel-Grau, M. (2023): "Employment and Reallocation Effects of Higher Minimum Wages," Working Paper, LMU Munich.

Du, P. and S. Wang (2020): "The Effect of Minimum Wage on Firm Markup: Evidence from China," *Economic Modelling*, 86, 241–250.

Dustmann, C., A. Lindner, U. Schönberg, M. Umkehrer, and P. vom Berge (2022): "Reallocation Effects of the Minimum Wage," *Quarterly Journal of Economics*, 137, 267–328.

Engbom, N. and C. Moser (2022): "Earnings Inequality and the Minimum Wage: Evidence from Brazil," *American Economic Review*, 112, 3803–3847.

Haanwinckel, D. (2025): "Supply, Demand, Institutions, and Firms: A Theory of Labor Market Sorting and the Wage Distribution," NBER Working Paper 31318.

Haelbig, M., M. Mertens, and S. Müller (2023): "Minimum Wages, Productivity, and Reallocation," IZA Discussion Paper 16160.

Harasztosi, P. and A. Lindner (2019): "Who Pays for the Minimum Wage?" *American Economic Review*, 109, 2693–2727.

Hau, H., Y. Huang, and G. Wang (2020): "Firm Response to Competitive Shocks: Evidence from China's Minimum Wage Policy," *Review of Economic Studies*, 87, 2639–2671.

Haucap, J., U. Pauly, and C. Wey (2001): "The Incentives of Employers' Associations to Raise Rivals' Costs in the Presence of Collective Bargaining," *International Journal of Industrial Organization*, 19, 1589–1618.

# References II


Luca, D. L. and M. Luca (2018): "Survival of the Fittest: The Impact of the Minimum Wage on Firm Exit," Harvard Business School NOM Unit Working Paper 17-088.

Mayneris, F., S. Poncet, and T. Zhang (2018): "Improving or Disappearing: Firm-Level Adjustments to Minimum Wages in China," *Journal of Development Economics*, 135, 20–42.

Rao, N. and M. Risch (2026): "Who's Afraid of the Minimum Wage?" *Quarterly Journal of Economics*, 141, 373–423.

Riley, R. and C. Rosazza Bondibene (2017): "Raising the Standard: Minimum Wages and Firm Productivity," *Labour Economics*, 44, 27–50.

Williamson, O. E. (1968): "Wage Rates as a Barrier to Entry: The Pennington Case in Perspective," *Quarterly Journal of Economics*, 82, 85–116.